

Engine

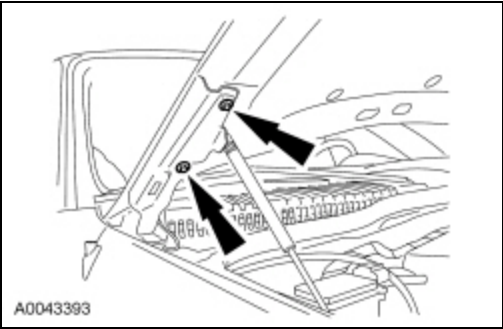
 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073) or equivalent
 ST3110-A	Support Brackets, Engine 303-1507 or equivalent

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. **NOTE:** *Index-mark the hood hinge location to aid in hood installation.*

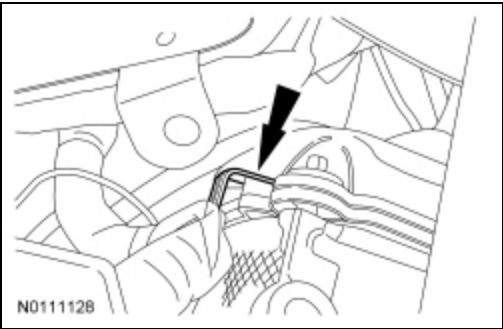
Remove the 4 bolts and the hood.



3. **NOTICE:** If the engine is repaired or replaced because of upper engine failure, typically including valve or piston damage, check the intake manifold for metal debris. If metal debris is found, install a new intake manifold. Failure to follow these instructions can result in engine damage.

Remove the intake manifold. For additional information, refer to Intake Manifold in this section.

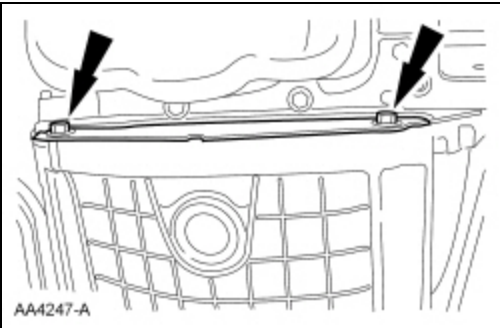
4. Detach the wiring harness pin-type retainer from the rear of the RH valve cover.



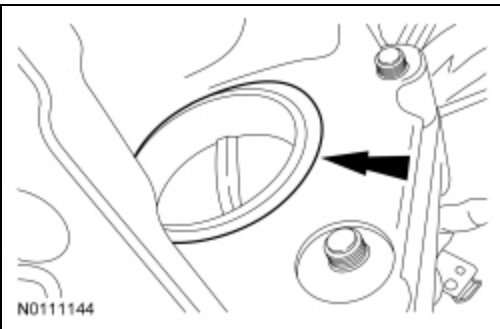
5. Remove the accessory drive belt.
6. Remove the cooling module. For additional information, refer to Section 303-03.
7. Position the Power Distribution Box (PDB) and wiring harness forward.
8. Remove the starter. For additional information, refer to Section 303-06.
9. Detach the starter motor wiring harness pin-type retainer from the oil pan.



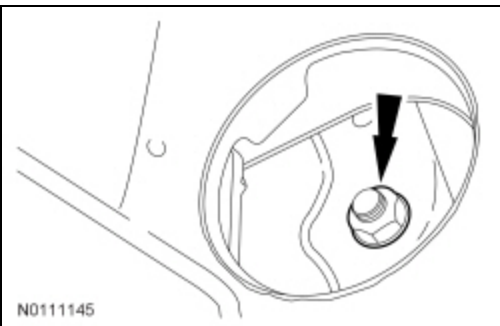
10. Remove the 2 bolts and the flexplate inspection cover.



11. Remove the cylinder block opening cover.

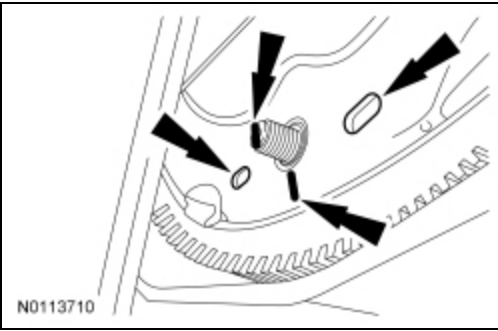


12. Remove the 4 torque converter-to-flexplate nuts.
 - Discard the nuts.



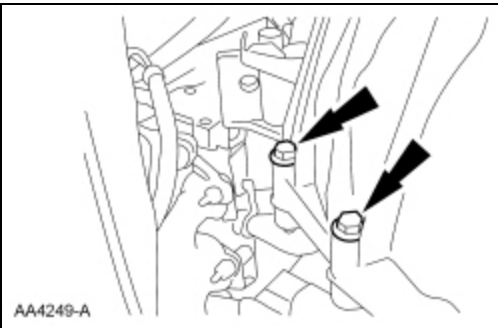
13. **NOTICE:** The torque converter must be installed in it's original orientation to the flexplate or engine damage may occur.

Rotate the flexplate so that the 2 slotted holes are shown at the 6 o'clock position and index mark the torque converter stud to the flexplate to maintain alignment during installation.



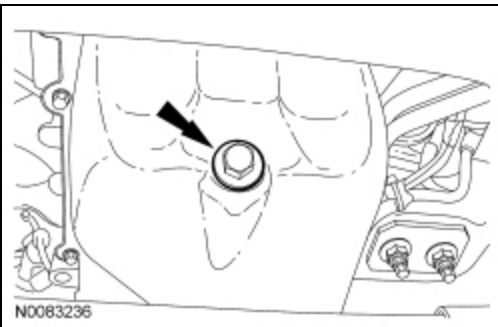
14. **NOTE:** *The upper 2 transmission-to-engine bolts will be removed later.*

Remove the lower 5 transmission-to-engine bolts.

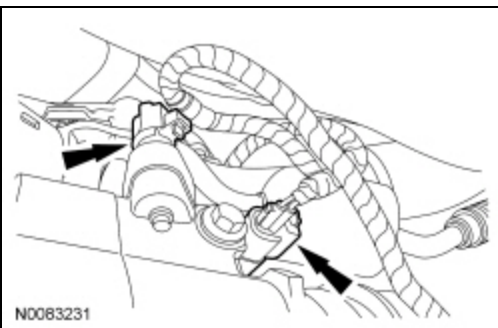


15. Remove the drain plug and drain the engine oil. Install the drain plug when finished.

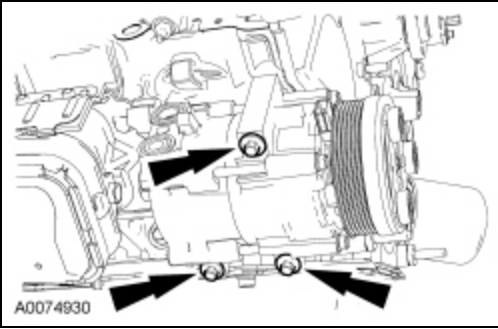
- Tighten to 23 Nm (17 lb-ft).



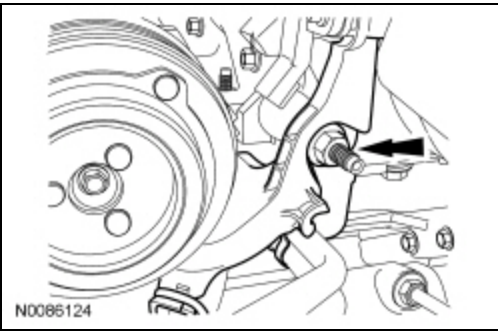
16. Disconnect the A/C compressor electrical connector and the wiring harness retainer.



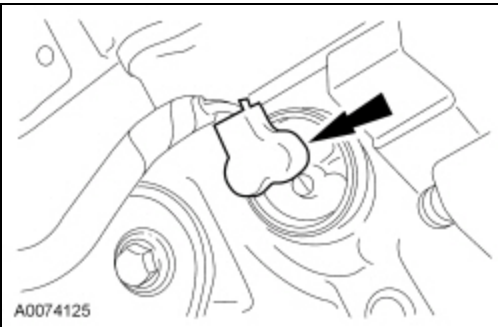
17. Remove the 3 bolts and position the A/C compressor aside.



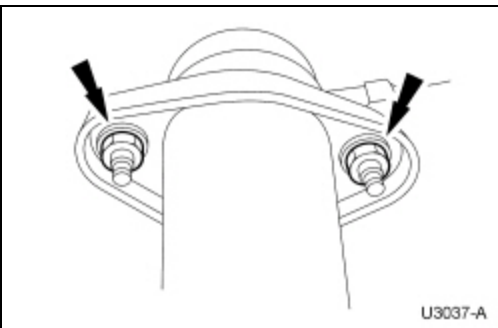
18. Remove the nut and the starter wiring harness retainer push pin and position aside the starter wiring harness and transmission cooler tube support bracket.



19. If equipped, disconnect the block heater electrical connector.



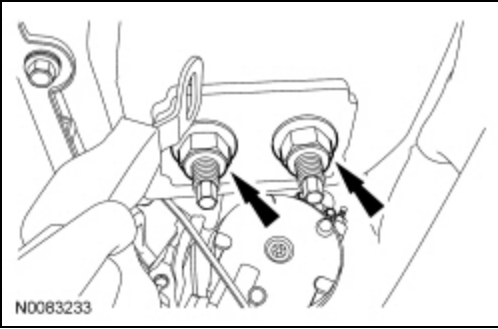
20. Remove the 4 exhaust Y-pipe flange nuts.



21. **NOTICE: Only use hand tools when removing the engine support insulator nuts or the engine support insulator may be damaged.**

NOTE: *If during nut removal the stud bolt is extracted from the engine support insulator, separate the nut from the stud bolt prior to stud bolt installation.*

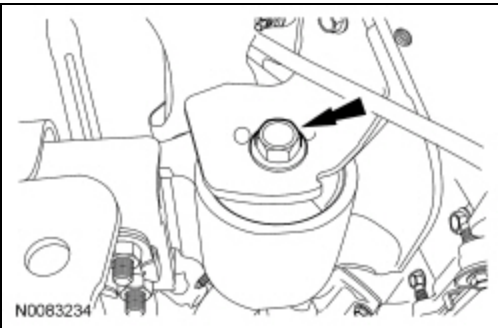
Remove the 2 RH engine support insulator nuts and studs.



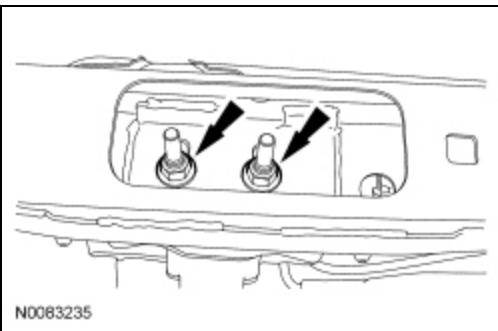
22. If equipped, remove the front drive shaft. For additional information, refer to Section 205-01.

23. **NOTICE: Only use hand tools when loosening the engine support insulator through bolt or the engine support insulator may be damaged.**

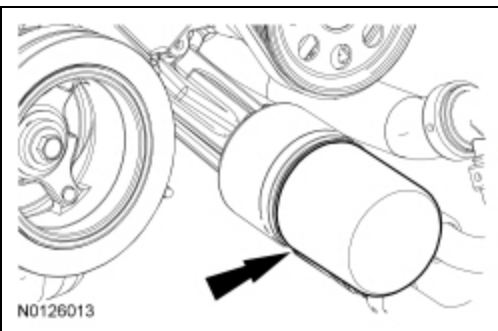
Remove the LH engine support insulator bolt.



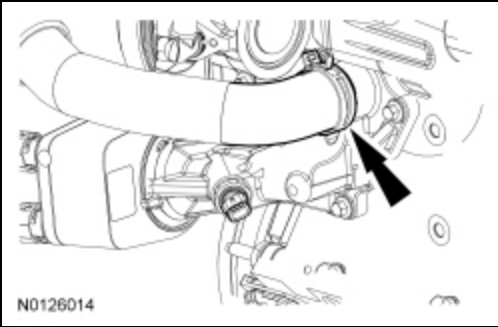
24. Loosen the 2 transmission mount nuts.



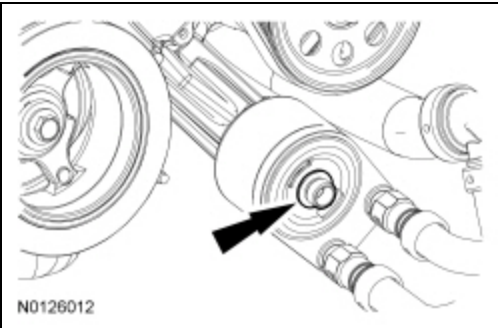
25. Remove and discard the oil filter.



26. Disconnect the lower radiator hose from the oil filter.

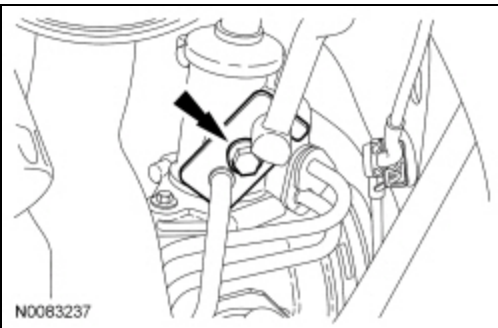


27. Remove the oil cooler threaded shaft and position aside the oil cooler assembly.



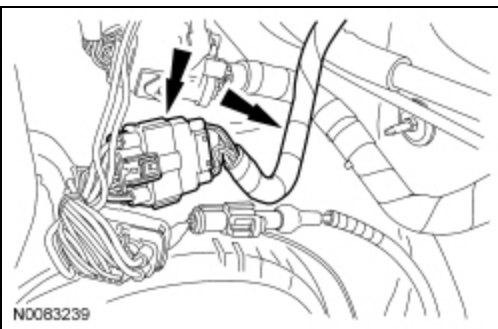
28. **NOTICE:** While servicing the power steering system, care should be taken to prevent the entry of foreign material or failure of the power steering components may result.

Remove the bolt and detach the power steering return hose from the steering gear.

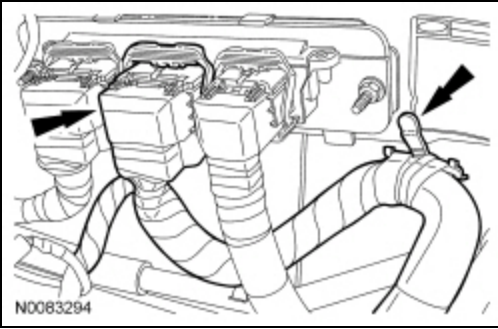


29. Remove the power steering pump. For additional information, refer to Section 211-02

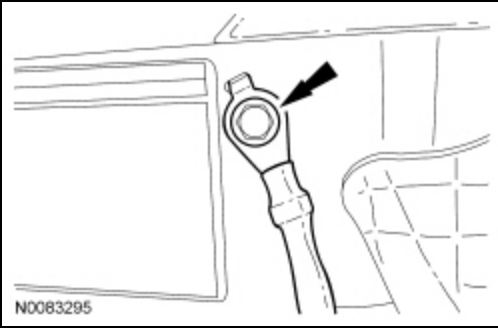
30. Disconnect electrical connector and detach the connector and wiring harness retainers.



31. Disconnect the PCM electrical connector and the engine wiring harness retainer and position the engine wiring harness aside.



32. Remove the ground strap bolt.

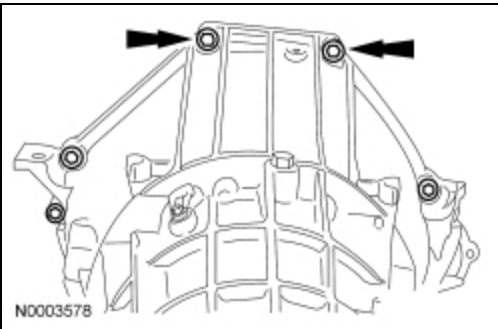


33. Disconnect the 2 heater hoses.

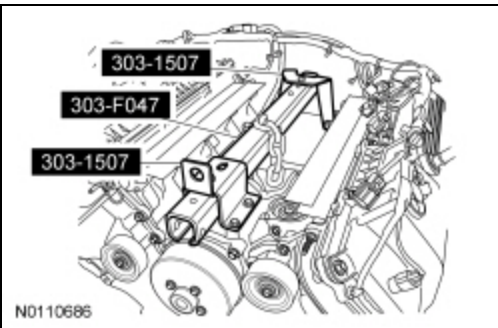
34. Support the transmission.

35. **NOTE:** *On Four-Wheel Drive (4WD), it may be necessary to reposition the transfer case vent hose to access the bolts.*

Remove the upper 2 transmission-to-engine bolts.



36. Install the Engine Lifting Bracket and Engine Support Brackets.



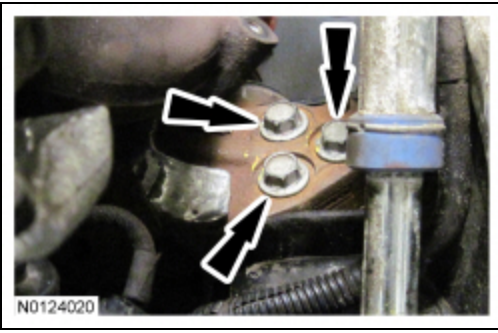
37. Remove the 2 bolts and the coolant expansion tank.

38. Remove the steering column shaft-to-steering gear bolt and disconnect the steering column shaft.

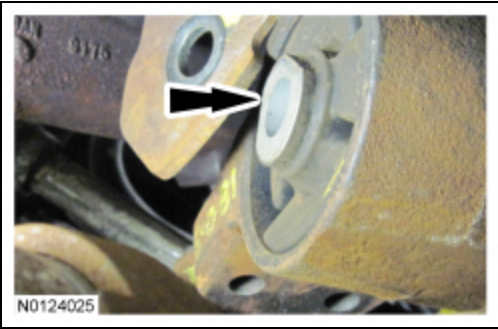
39. Using the Heavy Duty Floor Crane and the Engine Lifting Bracket, raise the engine assembly to gain access to the 3 LH engine support insulator bolts.

40. Remove the 2 bolts and the LH exhaust manifold heat shield.

41. Remove the 3 LH engine support Insulator bolts.



42. Remove the LH engine support insulator from under the vehicle.



43. Using a floor crane, remove the engine assembly from the vehicle.